



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
30-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

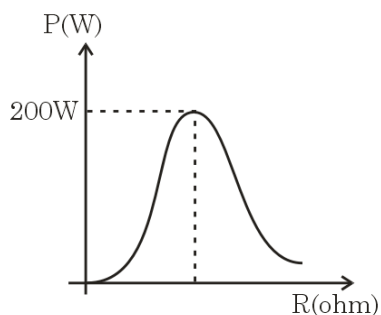
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

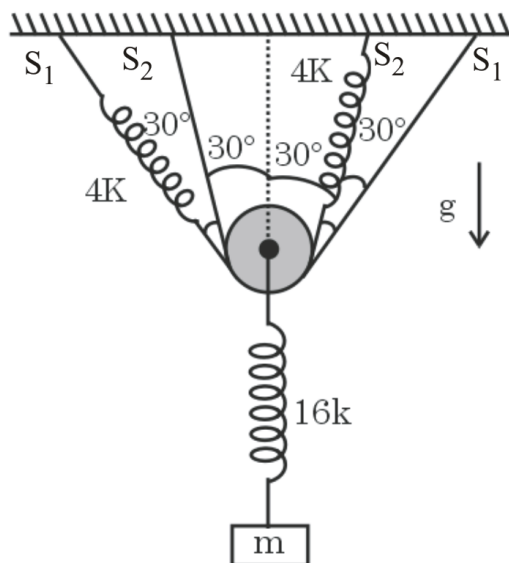
1. Moment of inertia of a uniform equilateral triangular lamina of mass 20 kg and side length 75 cm about an axis perpendicular to plane and passing through centroid (in kg-m^2) is :-
 (A) $\frac{360}{96}$ (B) $\frac{360}{32}$ (C) $\frac{360}{192}$ (D) $\frac{180}{192}$
2. A battery of emf 48 V unknown internal resistance is connected to a rheostat whose resistance can be varied in a long range. A graph is plotted between power developed across rheostat and resistance of rheostat which is shown in figure. The internal resistance of the battery in Ω is :-



- (A) 1.44 (B) 2.88 (C) 5.76 (D) 11.52

Space for Rough Work

3. All the springs and pulleys are ideal and massless. If the block is slightly pulled vertically downward from its equilibrium position, then the time period of oscillation of the block is (assume strings are taught all the time)



(A) $2\pi\sqrt{\frac{m}{4k}}$

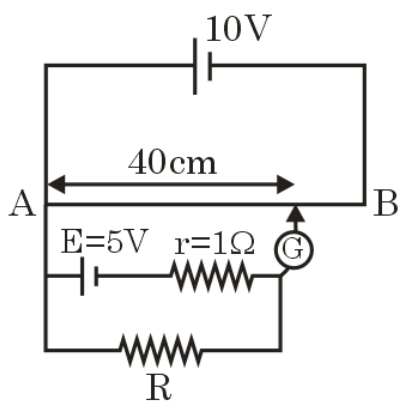
(B) $\pi\sqrt{\frac{2m}{k}}$

(C) $\pi\sqrt{\frac{m}{2k}}$

(D) $2\pi\sqrt{\frac{m}{16k}}$

Space for Rough Work

4. A potentiometer wire AB is 100 cm long and has total resistance of 10Ω . Find the value of unknown resistance R so that null point is obtained at a distance 40 cm from A.



(A) 1Ω

(B) 2Ω

(C) 3Ω

(D) 4Ω

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

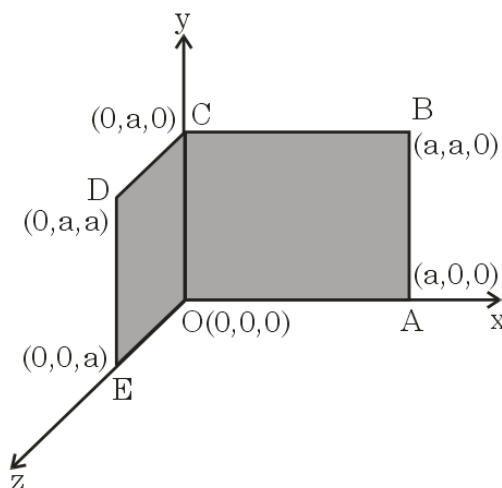
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

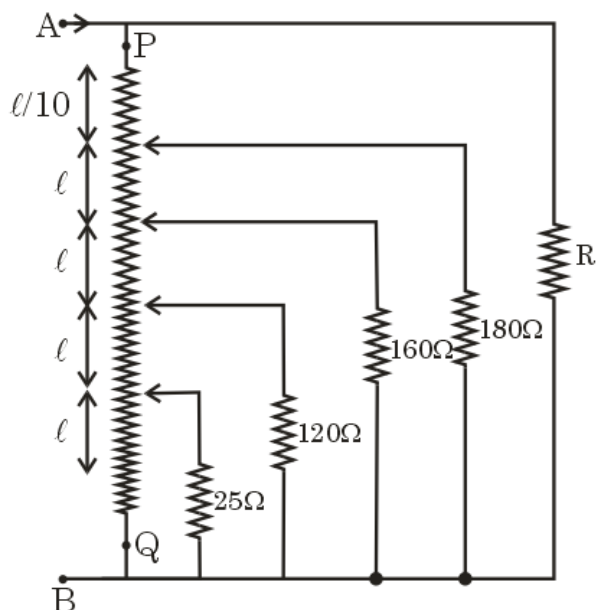
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Two surface OABC and OCDE lies in the plane of xy and yz as shown in the figure. A charged particle 'q' lies in the space at a point P, if:



- (A) coordinates of 'P' is $(a - \Delta r, a - \Delta r, \Delta r)$ and $a \gg \Delta r$, then flux passing through surface OABC is $\frac{7q}{24\epsilon_0}$.
- (B) coordinates of 'P' is $(a - \Delta r, a - \Delta r, \Delta r)$ and $a \gg \Delta r$, then flux passing through surface OCDE is $\frac{q}{24\epsilon_0}$.
- (C) coordinates of 'P' is $(a + \Delta r, a + \Delta r, -\Delta r)$ and $a \gg \Delta r$, then flux passing through surface OABC is $\frac{q}{24\epsilon_0}$.
- (D) coordinates of 'P' is $(a + \Delta r, a + \Delta r, -\Delta r)$ and $a \gg \Delta r$, then flux passing through surface OCDE is $\frac{q}{24\epsilon_0}$.

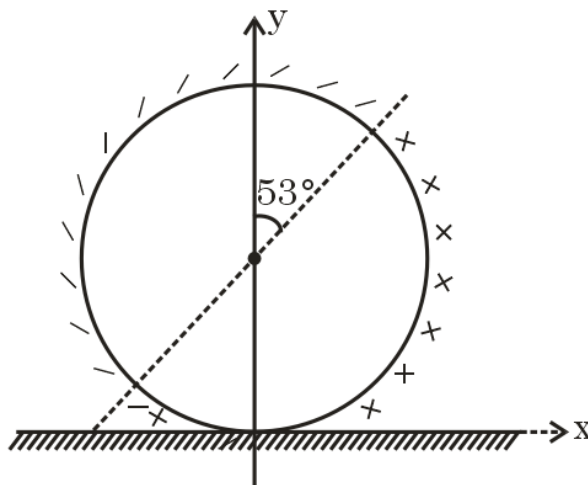
6. The resistance of wire PQ is $410\ \Omega$, which is tapped with the external wire as shown in the figure. The value of R is not known. If the potential difference between AB is 400 volts and the net current through the source battery is 6 ampere, then which of the following is/are correct :-



- (A) The resistance of unknown resistor R is $200\ \Omega$.
 (B) The equivalent resistance across AB is $\frac{200}{3}\ \Omega$.
 (C) The current in $25\ \Omega$ resistor is $\frac{2}{5}$ ampere.
 (D) The potential drop across $120\ \Omega$ resistor is 60 volts.

Space for Rough Work

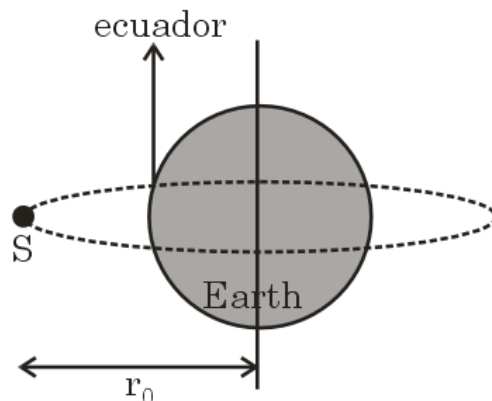
7. A non conducting uniform thin shell of mass m and radius R is having uniform charge density $+\sigma$ on one half and $-\sigma$ on another half as shown in figure. It is placed on a rough non conducting horizontal plane. At $t = 0$ a uniform electric field $\vec{E} = -E_0 \hat{j}$ N/C is switched on and the solid sphere starts rolling without slipping. Then which of the following is/are correct.



- (A) The acceleration of the topmost point of the hollow sphere at $t = 0$, is $\frac{36}{25} \frac{\pi \sigma E_0 R^2}{m}$.
- (B) The magnitude of frictional force at $t = 0$ is $\frac{18}{25} \pi \sigma E_0 R^2$.
- (C) The acceleration of centre of mass of the hollow sphere at $t = 0$, is $\frac{12}{25} \frac{\pi \sigma E_0 R^2}{m}$.
- (D) The magnitude of frictional force at $t = 0$ is $\frac{9}{25} \pi \sigma E_0 R^2$.

Space for Rough Work

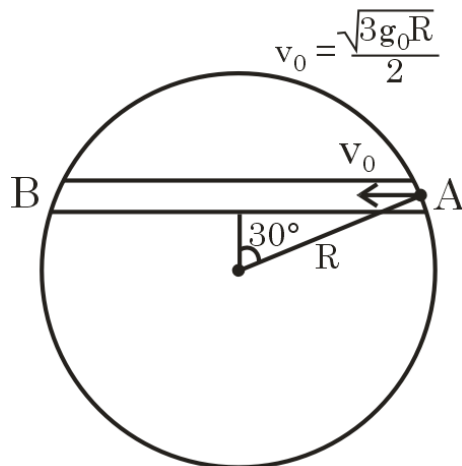
8. Consider a satellite 'S' which is just above the ECUADOR and rotating on equatorial plane of earth. The sense of rotation of earth and satellite are similar means both are rotating from west to east. The time period of satellite is 6 hours. At $t = 0$, satellite is just above the Brazil. Now choose the correct option (s).



- (A) The time after which the satellite will be above the ECUADOR again is 8 hours.
- (B) The above satellite may be a geostationary satellite.
- (C) The height of geostationary satellite above the sea level is 35,800 km approximately.
- (D) If the distance from the centre of earth to the satellite 'S' becomes $4r_0$, then the time after which the satellite will be above the ECUADOR again is 24 hours.

Space for Rough Work

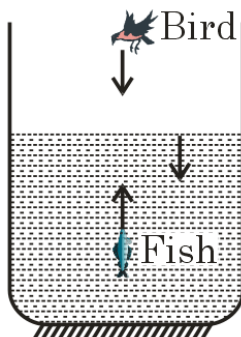
9. A tunnel is dug across the earth along a chord AB as shown, A particle is projected inside the tunnel from end 'A' with an initial velocity $v_0 = \frac{\sqrt{3g_0R}}{2}$, there is no friction between the tunnel and the particle. Then choose the correct option (s), (where g_0 = gravitational acceleration on the surface of Earth, R = radius of earth)



- (A) The time taken by the particle to move from end 'A' to other end 'B' of the tunnel is $\frac{\pi}{3} \sqrt{\frac{R}{g_0}}$.
- (B) The time taken by the particle to move from end 'A' to other end 'B' of the tunnel is $\frac{2\pi}{3} \sqrt{\frac{R}{g_0}}$.
- (C) The maximum speed of the particle inside the tunnel is $\sqrt{g_0R}$.
- (D) The maximum speed of the particle inside the tunnel is $\sqrt{2g_0R}$.

Space for Rough Work

10. A bird is flying downward with constant velocity 6 cm/s towards a fixed container filled with a liquid of refractive index $\frac{4}{3}$. A fish is moving upward with constant speed 4 cm/s towards the surface of liquid. Surface of liquid is also moving downward with constant velocity 2 cm/s shown in the figure. The bottom of container is silvered. Then :-



- (A) Speed of fish as seen by bird only after refraction is 9 cm/sec.
 (B) Speed of fish as seen by bird only after refraction is 8.5 cm/sec.
 (C) Speed of fish as seen by bird after reflection and then refraction is 3 cm/sec.
 (D) Speed of fish as seen by bird after reflection and then refraction is 2.5 cm/sec.

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

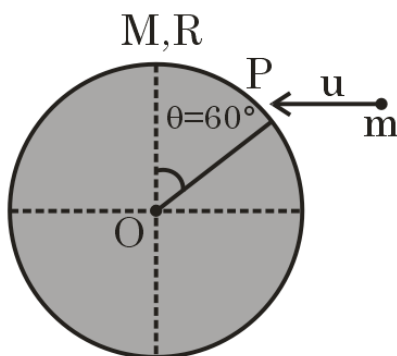
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A uniform circular disc of mass $M = 4\text{ kg}$ and radius $R = 25\text{ cm}$ is suspended in a vertical plane and hinged at its centre 'O'. It is free to rotate about a horizontal axis passing through its centre and perpendicular to its plane. A particle of mass $m = 0.5\text{ kg}$ moving horizontally with a velocity $u = 20\text{ m/s}$ collides with disc at point 'P' and gets stuck to it as shown. ($g = 10\text{ m/s}^2$)

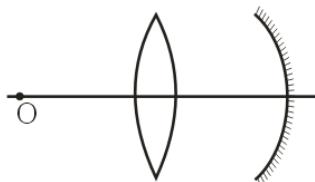


1. Find the angular velocity of the disc just after collision (in rad/sec)
2. The force on the disc due to hinge at its centre when the particle reaches the highest point (in newton).

Space for Rough Work

Paragraph for Questions 3 and 4

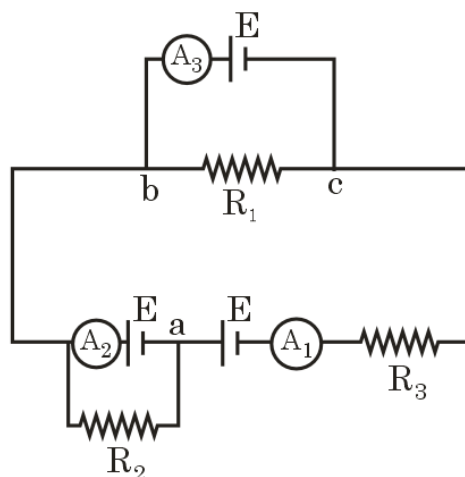
A converging lens of power $P_1 = +5D$ and a concave mirror of power $P_2 = +16D$ are placed at a separation 80 cm such that their principal axis coincide. A point object is placed left side of the lens on its principal axis at a distance X_1 from the lens such that image of the object coincides with the object itself. When the mirror is further shifted right side by a distance X_2 , the image of the object further coincides with the object.



3. The value of X_1 (in cm) is :
4. The value of X_2 (in cm) is:

Paragraph for Questions 5 and 6

In the circuit shown in figure, $R_1 = R_2 = R_3 = 3\Omega$ and e.m.f. of each cell is $E = 4V$ and negligible internal resistance. All ammeters are ideal.



5. Reading of ammeter A_1 in ampere is:
6. Potential difference (in Volt) between points a and b is

Space for Rough Work

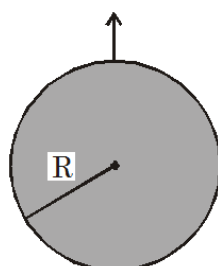
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

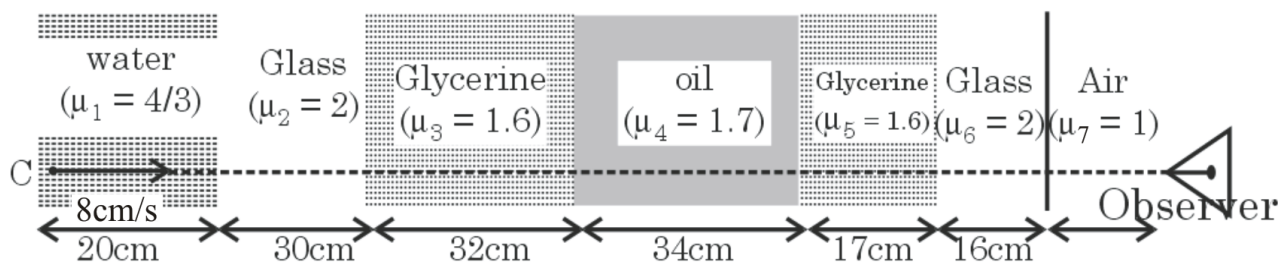
Zero Marks : 0 In all other cases

7. A small body of mass 'm' is projected from the surface of a given planet as shown in the figure. If small body has velocity which is just sufficient to get out from the gravity of planet then time taken by small body to reach at height of $3R$ from the surface of planet is $\frac{K}{3} \sqrt{\frac{2R^3}{GM}}$. Find the value of K.



Planet of mass M and radius R

8. A non-conducting solid sphere has a charge density given by $\rho = \rho_0 r^n$, where ρ_0 , n are constant and 'r' is the distance from the centre of sphere. If the magnitude of electric field at half of the radius is $\frac{1}{16}$ times that of the surfaces of nonconducting then find the value of n.
9. Find the velocity (in cm/s) of the car C as appeared to the observer shown in the figure. There are several parallel slabs of different medium placed between the car and the observer.



Space for Rough Work

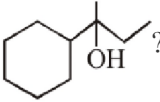
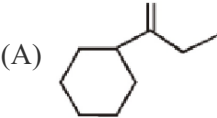
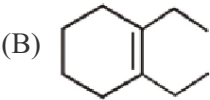
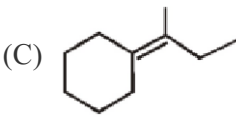
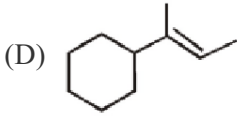
PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

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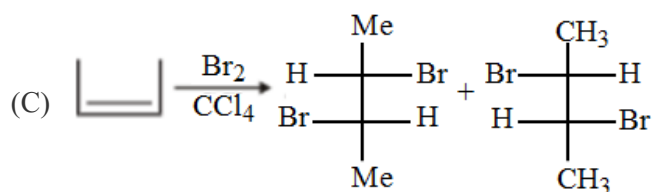
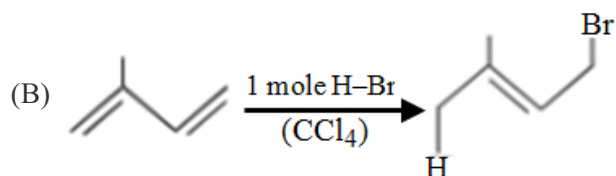
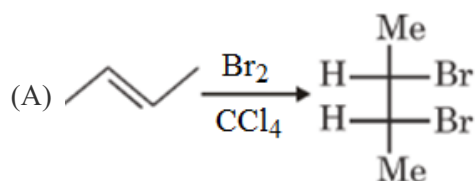
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Select the incorrect statement :
 - (A) MgFe_2O_4 can behave like a ferrimagnetic solid
 - (B) The unit cell having crystal parameters, $a = b \neq c$, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$ is hexagonal
 - (C) In ionic compounds having Frenkel defect, the ratio r^+/r^- is high
 - (D) The co-ordination number of Na^+ ion in NaCl is 6
2. If FeO crystallizes rock salt structure in which O^{2-} ions occupy FCC lattice points. Due to some crystal defect, lattice becomes $\text{Fe}_{0.75}\text{O}$. Find number of effective Fe^{3+} ions per unit cell -
 - (A) 1
 - (B) 2
 - (C) 3
 - (D) 4
3. Which of the following is not product of acidic dehydration of  ?
 - (A) 
 - (B) 
 - (C) 
 - (D) 

Space for Rough Work

4. Identify reactions correctly matched with their major product ?



(D) All of these

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. The CORRECT order(s) regarding the atomic size is / are

(A) $Zr \simeq Hf$

(B) $Fe \simeq Co \simeq Ni$

(C) $Sc < Y < La$

(D) $Cu < Zn$

6. Which of the following compound(s) give(s) dibasic acid as a product on hydrolysis.

(A) P_4O_6

(B) SO_2Cl_2

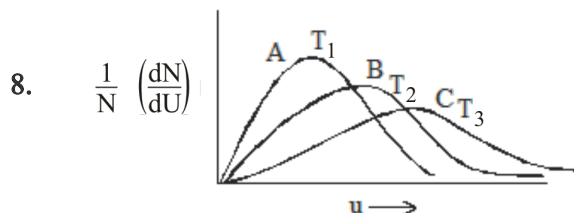
(C) SF_6

(D) SeF_6

Space for Rough Work

7. Which of the following ore(s) contain(s) copper metal?

- (A) Dolomite
- (B) Chalcopryrite
- (C) Fool's gold
- (D) Malachite



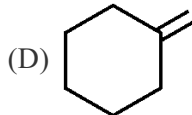
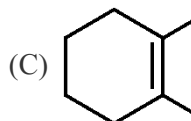
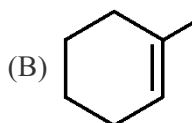
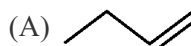
Which of the following statement is correct.

- (i) If A & B are same gases then $T_2 > T_1$
- (ii) If $T_1 = T_2$ then $A_{(g)}$ may be $SO_{2(g)}$ & $B_{(g)}$ may be $CH_{4(g)}$
- (iii) If B & C are same gases then $T_3 > T_2$
- (iv) If $T_1 = T_2 = T_3$ then $A_{(g)}$ may be CH_4 , $B_{(g)}$ may be SO_2 & $C_{(g)}$ may be $SO_{3(g)}$

- (A) i, ii & iii only
- (B) i, ii, & iv only
- (C) All are correct
- (D) ii & iii only

Space for Rough Work

9. Two volatile liquids A & B are mixed to form an ideal solution. If $P_A^0 = 100$ mm of Hg & $P_B^0 = 150$ mm of Hg then identify the correct option(s) if 10 moles of A are mixed with 40 moles of B such that moles vaporised are very small.
- (A) Total vapour pressure will be 140 mm of Hg.
- (B) Mole fraction of A in vapour state will be less than 0.5.
- (C) Mole fraction of B in vapour state is equal to $\frac{6}{7}$
- (D) Mole % of B in vapour state will be greater than 80%.
10. Markownikoff's rule is / are applicable on which of the following alkene :



Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
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Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

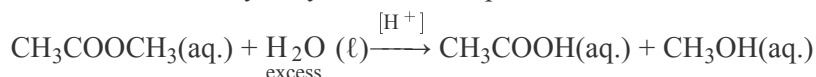
Paragraph for Questions 1 and 2

The complex compounds show space isomerism when the ligand species have different spatial arrangement surrounding the CMA

1. What is the number of stereo isomeric forms for the complex $[\text{Pt}(\text{py})(\text{NH}_3)(\text{Br})(\text{I})]$
2. How many stereo isomers are possible for $[\text{Cr}(\text{H}_2\text{O})_2(\text{C}_2\text{O}_4)_2]^-$?

Paragraph for Questions 3 and 4

Kinetics of acidic hydrolysis of ester is pseudo I order reaction



Rate law is given by : $r = K[\text{H}^+][\text{H}_2\text{O}][\text{CH}_3\text{COOCH}_3]$; $K = 1.8 \times 10^{-3} \text{ M}^{-2} \text{ sec}^{-1}$

where H^+ ion concentration is given by acid catalyst. In an experiment 1M $\text{CH}_3\text{COOCH}_3$ is hydrolysed using 0.1M HCl as catalyst. [Use molarity of $\text{H}_2\text{O} = \frac{1000}{18} \text{ M}$]

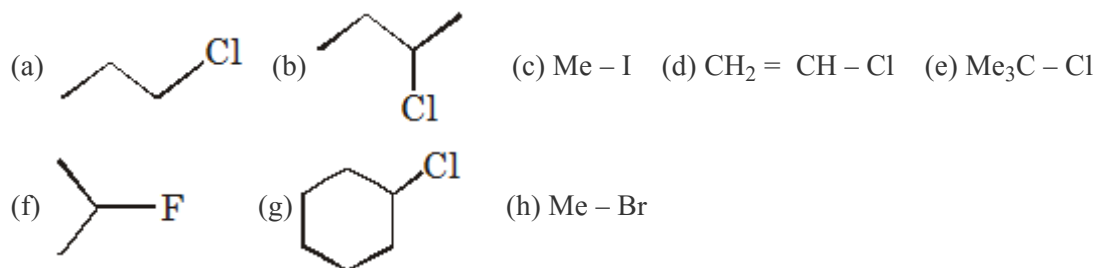
3. Reaction mixture is titrated against standard NaOH solution at different times. If initially 10 ml of NaOH was required while after long time 100 ml of NaOH is required then find volume of NaOH (in ml) required at 230.3 sec.
4. Calculate the time at which ester concentration reduce to 0.25M (in seconds) [$\ln 2 = 0.7$]

Space for Rough Work

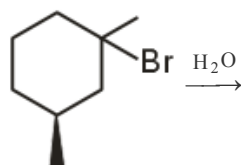
Paragraph for Questions 5 and 6

Paul Walden the brilliant son of a farmer succeeded in finding a concept that is now widely known as umbrella effect in S_N^2 reactions, in an S_N^2 reaction, the configuration at the reaction centre is inverted as an umbrella turns inside out in a strong wind. An ideal S_N^1 reaction, on the other hand, yields nearly racemised product as it proceeds through planar carbocation. S_N^2 reactions proceed smoothly on primary alkyl halides, whereas tertiary halides are fit for S_N^1 mechanism. However, evidences indicate that all the molecules of particular substrates do not necessarily react by a single mechanism. After all, the intention of the molecules is very clear, they want to follow the low energy path. If both paths have nearly the same energy barrier, they follow a mixed mechanism, for example, secondary alkyl halides follow a mixed mechanism, say 60% molecules react by S_N^2 and rest 40% by S_N^1 . Stereochemical features, as mentioned above, are noticeable only when the reaction is carried out on a particular enantiomer. Even the reaction centers, which are parts of suitably substituted rings, also exemplify the said features.

5. How many of the following are less reactive than Me-Cl in an S_N2 mechanism reaction



6. Total number of possible product(s) in given reaction?



Space for Rough Work

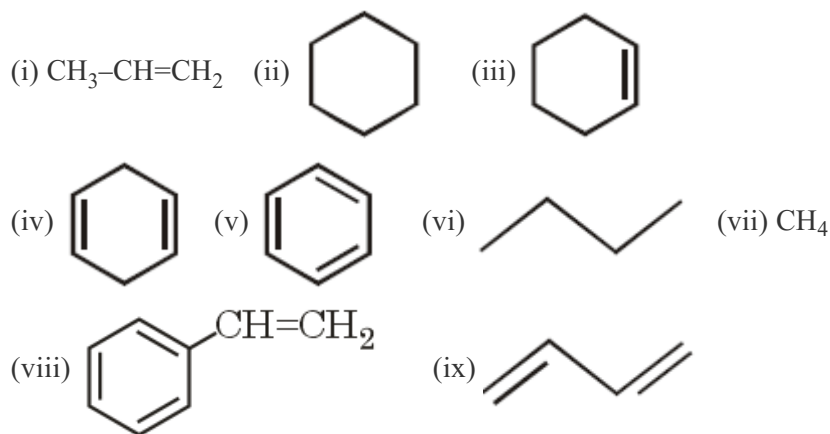
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Total number of complex compound which have low spin as well as Diamagnetic.
 $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{NiF}_6]^{2-}$, $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{Ni}(\text{en})_3]^{3+}$, $[\text{PtCl}_4]^{2-}$
8. Total number of species in which highest occupied molecular orbital has gerade type symmetry.
 B_2 , N_2 , C_2 , N_2^{2+} , O_2 , O_2^{2-} , N_2^{2-}
9. How many of the following will decolourise bromine water solution :



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Equation of normal to the curve $y = x \sin x$ at $x = \frac{\pi}{2}$ makes a triangle with coordinate axis of area A, then value of $[A]$ is equal to (Where $[.]$ denotes greatest integer function)

- | | |
|-------|-------|
| (A) 5 | (B) 4 |
| (C) 6 | (D) 2 |

2. Value of $\left[\int_0^{\pi/2} \sin^4 x \, dx \right]$ is

(Where $[.]$ denotes greatest integer function)

- | | |
|-------|-------|
| (A) 0 | (B) 1 |
| (C) 3 | (D) 4 |

Space for Rough Work

3. Let f be a non-negative function in $[0,1]$ and twice differentiable in $(0,1)$.

If $\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt$, $0 \leq x \leq 1$ and $f(0) = 0$, then $\lim_{x \rightarrow 0} \frac{1}{x^2} \int_0^x f(t) dt$:

- (A) equals 0 (B) equals 1
(C) does not exist (D) equals $\frac{1}{2}$

4. If $I_1 = \int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x} + \sqrt{\tan x}} dx$ & $I_2 = \int_{-\pi}^{\pi} \frac{\cos^2 x}{1 + a^x} dx$, then-

- (A) $\frac{1}{I_1} - \frac{1}{I_2} = \frac{3}{\pi}$
(B) $\frac{2}{I_1} - \frac{1}{I_2} = \frac{5}{\pi}$
(C) $\frac{2}{I_1} + \frac{1}{I_2} = \frac{8}{\pi}$
(D) $\frac{1}{I_1} + \frac{1}{I_2} = \frac{6}{\pi}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. From the point (1, 2), tangents are drawn to the curve $y^2 - 2x^3 - 4y + 8 = 0$. Then

- (A) sum of x-coordinates of points of contact is zero
 - (B) sum of x-coordinates of points of contact is 4
 - (C) sum of y-coordinates of points of contact is zero
 - (D) sum of y-coordinates of points of contact is 4
-

Space for Rough Work

6. Consider two family of straight lines
 $F_1 : 2x - y - 9 + \lambda (3x + y - 21) = 0$ and $F_2 : 2x + y - 5 + \mu (4x - y - 1) = 0$ where λ & μ are parameters. If L_1, L_2 are two lines belonging to F_1 and F_2 respectively, then identify which of the following statement(s) is(are) correct?
- (A) If L_1 & L_2 are parallel, then maximum distance between them is 5.
 (B) If L_1 & L_2 are parallel, then maximum distance between them is 3.
 (C) If L_1 & L_2 are perpendicular, then locus of their points of intersection is $(x - 1)(x - 6) + (y - 3)^2 = 0$.
 (D) If L_1 & L_2 are perpendicular, then locus of their points of intersection is $x^2 + y^2 - 6x + 6y + 9 = 0$.
7. If $f(x) = \sin^{-1} \sqrt{\frac{\sqrt{1+x^2} - 1}{2\sqrt{1+x^2}}}$, then which of the following is(are) correct?
- (A) $f'(-1) = \frac{-1}{4}$
 (B) Range of $f(x)$ is $\left[0, \frac{\pi}{2}\right]$
 (C) $f'(x)$ is an odd function
 (D) $\lim_{x \rightarrow 0} \frac{f(x)}{x} = \frac{1}{2}$

Space for Rough Work

8. If $z = \sec^{-1} \left(x + \frac{1}{x} \right) + \sec^{-1} \left(y + \frac{1}{y} \right)$, where $xy < 0$, then the possible values of z is (are)

(A) $\frac{8\pi}{10}$

(B) $\frac{7\pi}{10}$

(C) $\frac{9\pi}{10}$

(D) $\frac{21\pi}{20}$

9. Consider the function $f(x) = \lim_{n \rightarrow \infty} \frac{(1 + \cos x)^n + 5 \log_e x}{2 + (1 + \cos x)^n}$, then

(A) $f(x)$ is continuous at positive odd integral multiple of π

(B) $f(x)$ is discontinuous at positive even integral multiple of π

(C) $f(x)$ is discontinuous at positive odd integral multiple of $\frac{\pi}{2}$

(D) $f(x)$ is continuous at positive even integral multiples of $\frac{\pi}{2}$

10. $\int \left(1 + \tan x \tan \frac{x}{2} \right) dx$ is equal to-
(where C is constant of integration)

(A) $\int \left(\tan x \cot \frac{x}{2} - 1 \right) dx$

(B) $\int \sec x dx$

(C) $-\ln |\sec x - \tan x| + C$

(D) $-\ln \left| \frac{1 - \tan \frac{x}{2}}{1 + \tan \frac{x}{2}} \right| + C$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let $f(x)$ be a polynomial function satisfying the following conditions $\lim_{x \rightarrow \infty} \frac{f(x)}{|x|^3} = 0$, $\lim_{x \rightarrow \infty} (\sqrt{f(x)} - x) = -1$ and $f(0) = 0$ (Note : $[k]$ denotes greatest integer function less than or equal to k)

On the basis of above information, answer the following questions :

1. The number of points of discontinuity of $g(x) = [f(x)]$ in $[0, 3]$ is
2. The number of points of non-derivability of $h(x) = |f(|x|)|$ is -

Paragraph for Questions 3 and 4

Consider $f : \mathbb{N} \rightarrow \mathbb{N}$, where $f(n) = \frac{4}{\pi} \sum_{i=1}^n \sum_{j=1}^n \cot^{-1} \left(\frac{i}{j} \right)$

3. Value of $f(5) + f(10)$ is -
4. $\sum_{r=1}^{10} f(r)$, $r \in \mathbb{N}$ is

Space for Rough Work

Paragraph for Questions 5 and 6

If $f(x)$ be an even periodic function with period 6 defined as : $f(x) = \begin{cases} \{x\} & ; x \in [0, 1) \\ x^2 - 2x + 2 & ; x \in [1, 2) \\ 6 - 2x & ; x \in [2, 3] \end{cases}$

5. $\int_{-3}^1 f(x) \, dx$ is :

6. If $\frac{\int_0^1 f(x) \, dx + \int_2^3 f(x) \, dx}{\int_1^2 f(x) \, dx} = \frac{m}{n}$ where m and n are coprime then $m + n$ is

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. ABCD is a rectangle. Equation of AC and BD are $2x + 5y = 7$ and $5x + 2y = 7$ respectively. If $(10, -11)$ lies on AB and $\left(\frac{1}{2}, \frac{9}{2}\right)$ lies on CD, then area of rectangle ABCD is 7λ , where λ is (slope of line AB is negative)
8. If $k = \int_{-2022}^{2022} \frac{dx}{1 + x^{2021} + \sqrt{1 + x^{4042}}}$ then the sum of digit of value k is
9. If $L = \lim_{n \rightarrow \infty} \left(\frac{n+2}{n^2 + n + 1} + \frac{n+4}{n^2 + 2n + 4} + \frac{n+6}{n^2 + 3n + 9} + \dots + \frac{5}{7n} \right)$, then e^L is equal to -
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work